Microsoft 365

Learning Path 7: Manage Exchange Hybrid Settings and Migrations



Learning Path Agenda



Explore Exchange Hybrid Deployment Requirements



Plan and Configure a Hybrid Deployment Using the Hybrid Configuration Wizard



Lab 7-Exercise 1 – Prepare Azure AD for a Hybrid Deployment



Plan Mailbox Migrations



Run Mailbox Migrations



Lab 7-Exercise 2 – Configure your Hybrid Deployment



Troubleshoot hybrid deployments

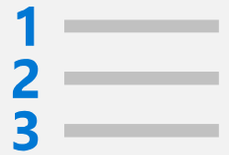


Lab 7-Exercise 3 – Test your Hybrid Deployment



Exercise – Planning a Mail Migration to Microsoft 365

Module 1: Explore Exchange Hybrid Deployment Requirements



Introduction

This Module examines the features, components, and requirements necessary to implement a hybrid deployment

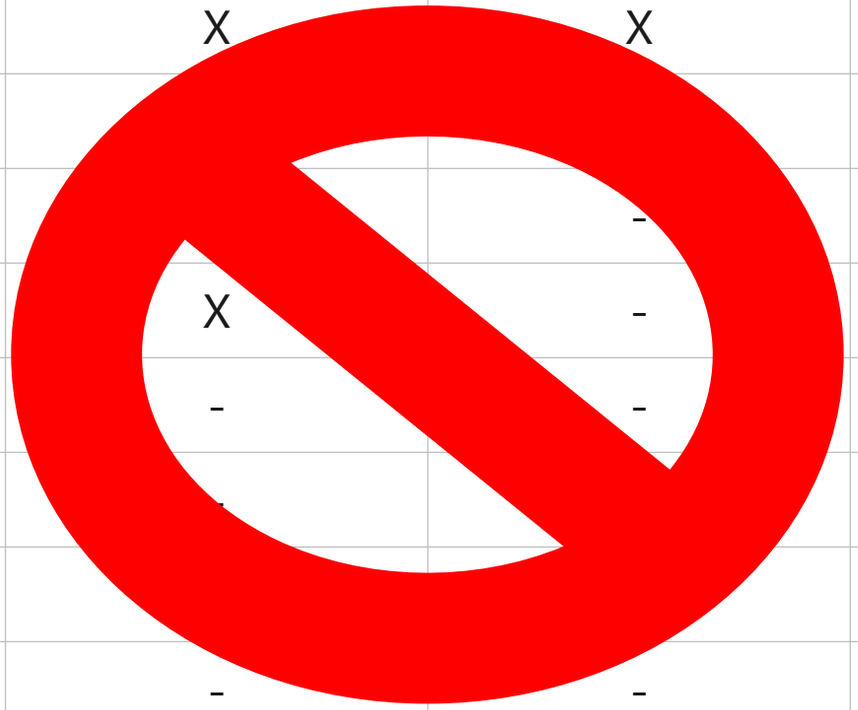
After this Module, you'll be able to:

- Describe connection options that are available for connecting on-premises Exchange to Microsoft 365
- List and describe components of a hybrid deployment
- Describe Azure Active Directory Connect (Azure AD Connect)
- Identify Microsoft 365 identity options for Exchange hybrid
- Compare Exchange delegated federation vs. OAuth



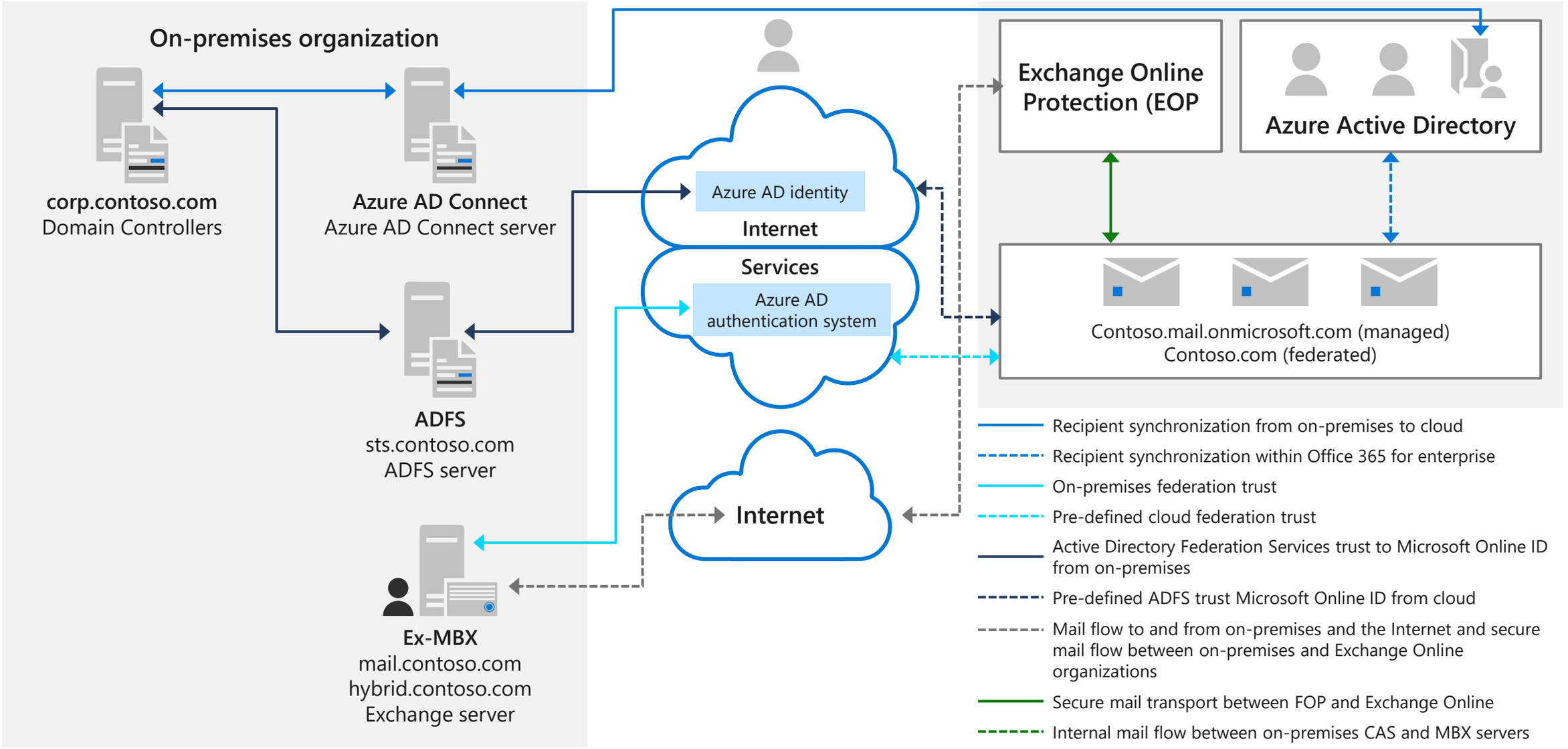
Examine the connection options for Exchange to Microsoft 365

	IMAP Migration	Cutover Migration	Staged Migration	Hybrid/Native
Exchange 5.5/2000	X	-	-	-
Exchange 2003	X	X	X	X*
Exchange 2007	X			X*
Exchange 2010	X		-	X
Exchange 2013-SE	X	X	-	X
Notes/Domino	X	-	-	-
GroupWise	X	-		-
GMAIL	X			X
Tenant to Tenant	-	-	-	X
Other	X	-	-	-



* Exchange 2003 or 2007 requires at least an Exchange 2010 Server for Hybrid

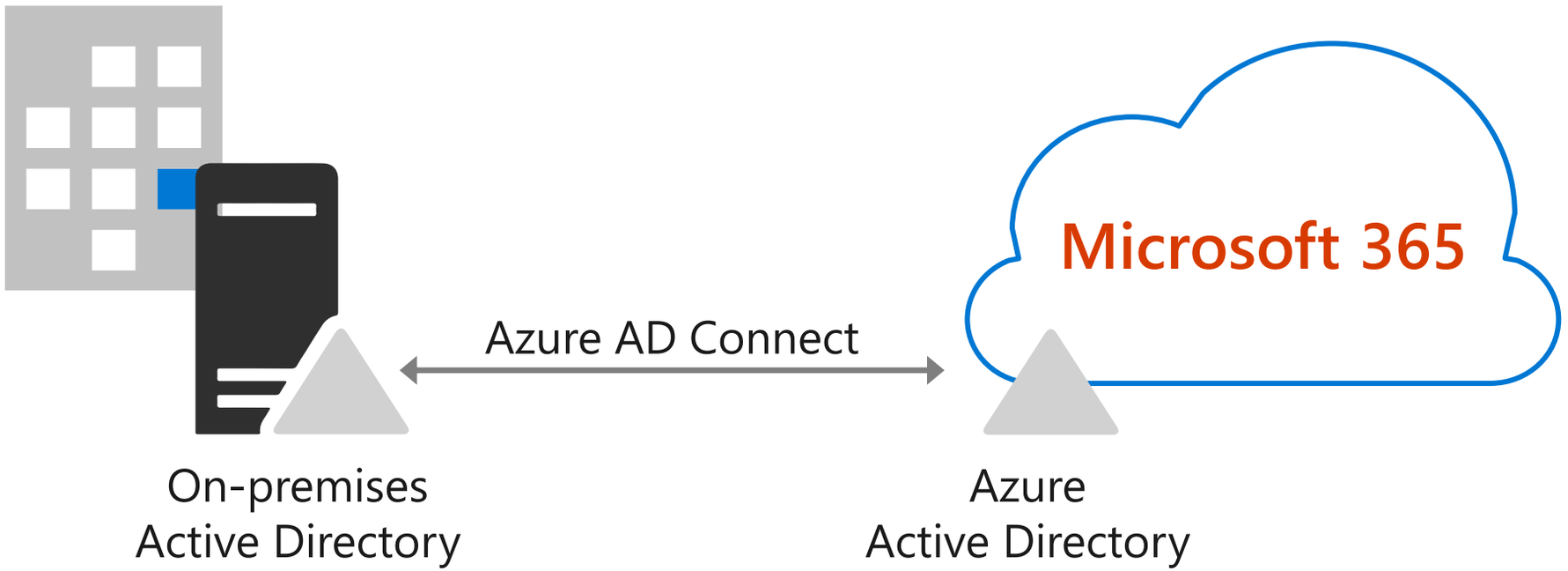
Explore the components of a hybrid deployment



Explore Azure Active Directory Connect

The Azure AD Connect tool, formerly known as DirSync, is the directory synchronization tool for Microsoft 365

The tool enables coexistence between an organization's on-premises Active Directory Domain Services environment (AD DS) and Microsoft 365 in the cloud



Examine the Microsoft 365 identity options for a hybrid deployment



Azure Active Directory (Azure AD) is an online instance similar to on-premises Active Directory

Azure AD provides authentication and authorization for Microsoft 365 and Microsoft 365 services, such as Windows devices registered with Azure AD, Intune, and Dynamics 365

Authentication options for Microsoft 365 identities into one of three main categories:

Cloud only - the user identity only exists in the cloud, so all password management and policy control are done through Azure AD



Directory synchronization with optional Password hash synchronization - two sets of security credentials - one on-premises and one in Azure AD. These credentials are kept aligned through directory synchronization and password synchronization

Directory synchronization with Pass-through authentication (PTA) or federated with AD FS – with federation, users are redirected to the company's sign-in page and authentication occurs using a delegated token. With PTA, there's no redirection; instead, Azure AD validates the credentials against the company's Active Directory



Seamless Single Sign-On (SSO) without AD FS

Seamless SSO allows domain-joined users to bypass prompts for passwords and authenticate using traditional intranet methods



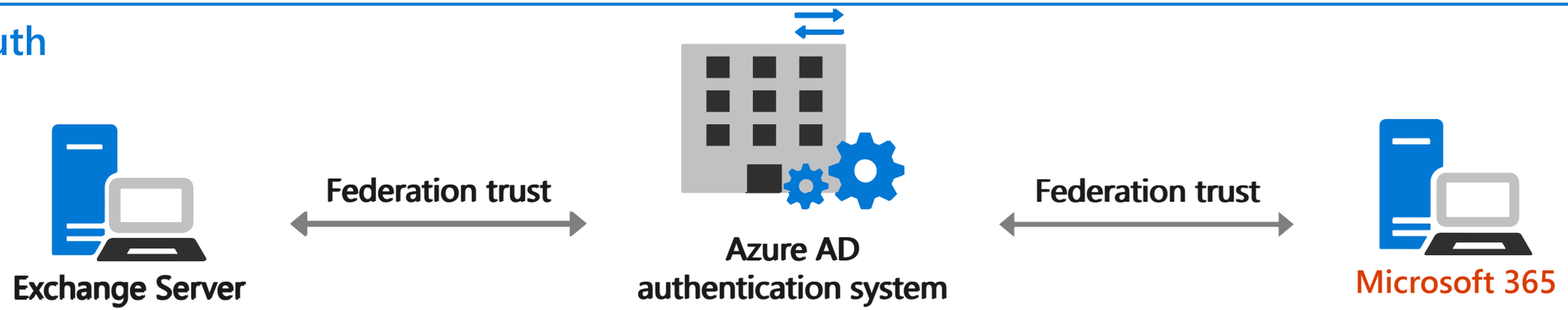
User account provisioning

Companies must consider on-premises, cloud, and hybrid scenarios

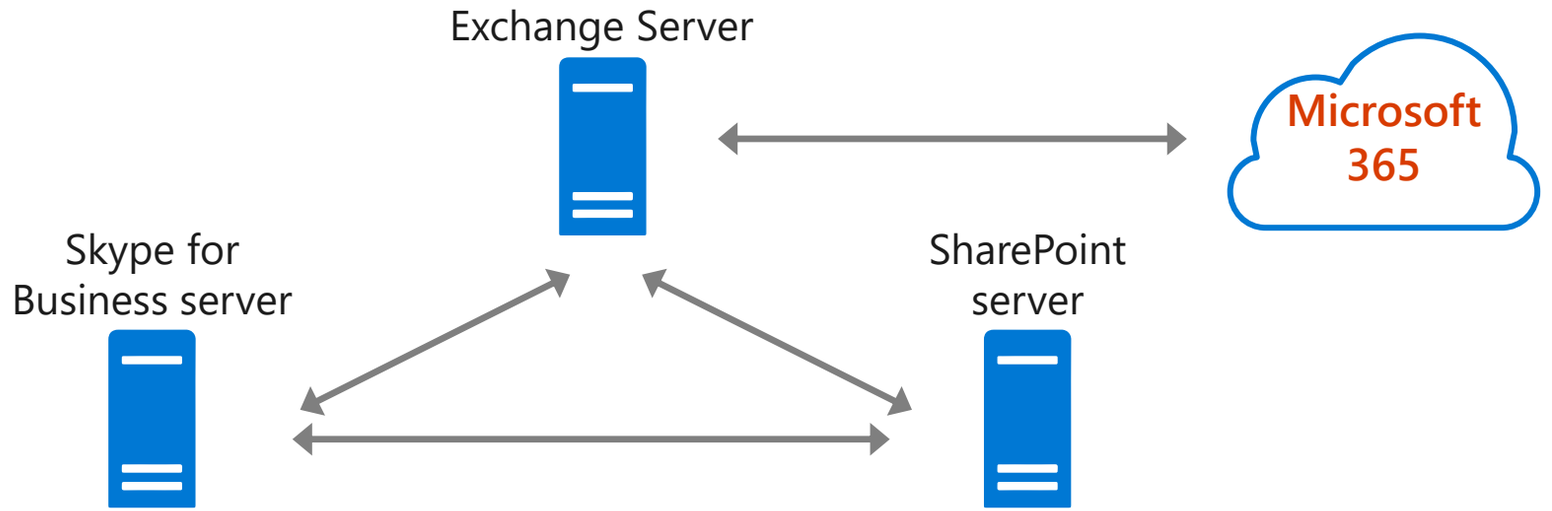
Compare Exchange delegated federation and Open Authentication

Two federation methods in Exchange Server:
Exchange delegated federation, otherwise known as Delegated Authentication (DAuth)
Open Authentication (OAuth), which was introduced with Exchange Server 2013

DAuth



OAuth



Knowledge Check

Test your knowledge by answering the Knowledge Check questions at the end of this Learn module



Summary

This Module examined the following items related to Exchange hybrid deployment requirements:



Organizations can choose one of the following migration strategies from on-premises Exchange Server to Microsoft 365:

IMAP migration
Staged migration
Cutover migration
Hybrid deployment



The Azure AD Connect tool, formerly known as DirSync, is the primary supported directory synchronization tool for Microsoft 365. It's made up of three parts:

Synchronization services
An optional authentication service
Azure AD Connect Health Monitoring service



Azure AD provides authentication and authorization for Microsoft 365 and Microsoft 365 services



Two federation methods in Exchange Server:

Exchange delegated federation, otherwise known as Delegated Authentication (DAuth)
Open Authentication (OAuth), which was introduced with Exchange Server 2013

LAB 4 – Übung 1 – Aufgabe 1-5

Module 2: Plan and Configure a Hybrid Deployment Using the Hybrid Configuration Wizard



Introduction

This Module examines best practices for configuring an Exchange hybrid deployment, as well as all planning requirements before running the Hybrid Configuration Wizard

After this Module, you'll be able to:

- Plan for Exchange Hybrid configuration
- Describe Organization Configuration Transfer
- Explain Exchange Modern Hybrid and Hybrid Agent
- Plan mail flow options for a hybrid deployment
- Plan to run the Hybrid Configuration Wizard



Plan for Exchange hybrid configuration



An Exchange hybrid configuration is a connection from a single or multiple on-premises Exchange organization to a single Exchange Online organization in a Microsoft 365 tenant

Exchange hybrid is commonly used for mid-term to long-term coexistence

Mailboxes are stored in both Exchange Server and Exchange Online for compliance and security purposes



The Hybrid Configuration Wizard (HCW) is the tool that builds the hybrid deployment environment



The HCW can configure two hybrid configuration options:

Minimal hybrid configuration - provides a quick way to move to Exchange Online, and a reasonable option for both the administrator and users

Full hybrid configuration - configures the full feature set to provide the best Exchange hybrid experience for your users



When should you use minimal and full hybrid configurations?

Examine Organization Configuration Transfer



The Organization Configuration Transfer (OCT) is a feature of the HCW

The OCT enables a company to complete a one-time transfer of its organizational settings from its Exchange Server on-premises environment to Exchange Online

After the transfer, you must make changes in both locations to keep the objects in sync



OCT supports Exchange Server 2010 and later



OCT requires the latest cumulative update or update roll-up available for the version of Exchange that’s installed in the on-premises organization



Additional OCT considerations include:

OCT can't match all settings, so always make sure you review them

If necessary, you can revert the OCT settings using the rollback script available in the logging folder of the HCW

Examine Exchange Modern Hybrid and the Hybrid Agent

A core component of Exchange Modern Hybrid is the Hybrid Agent

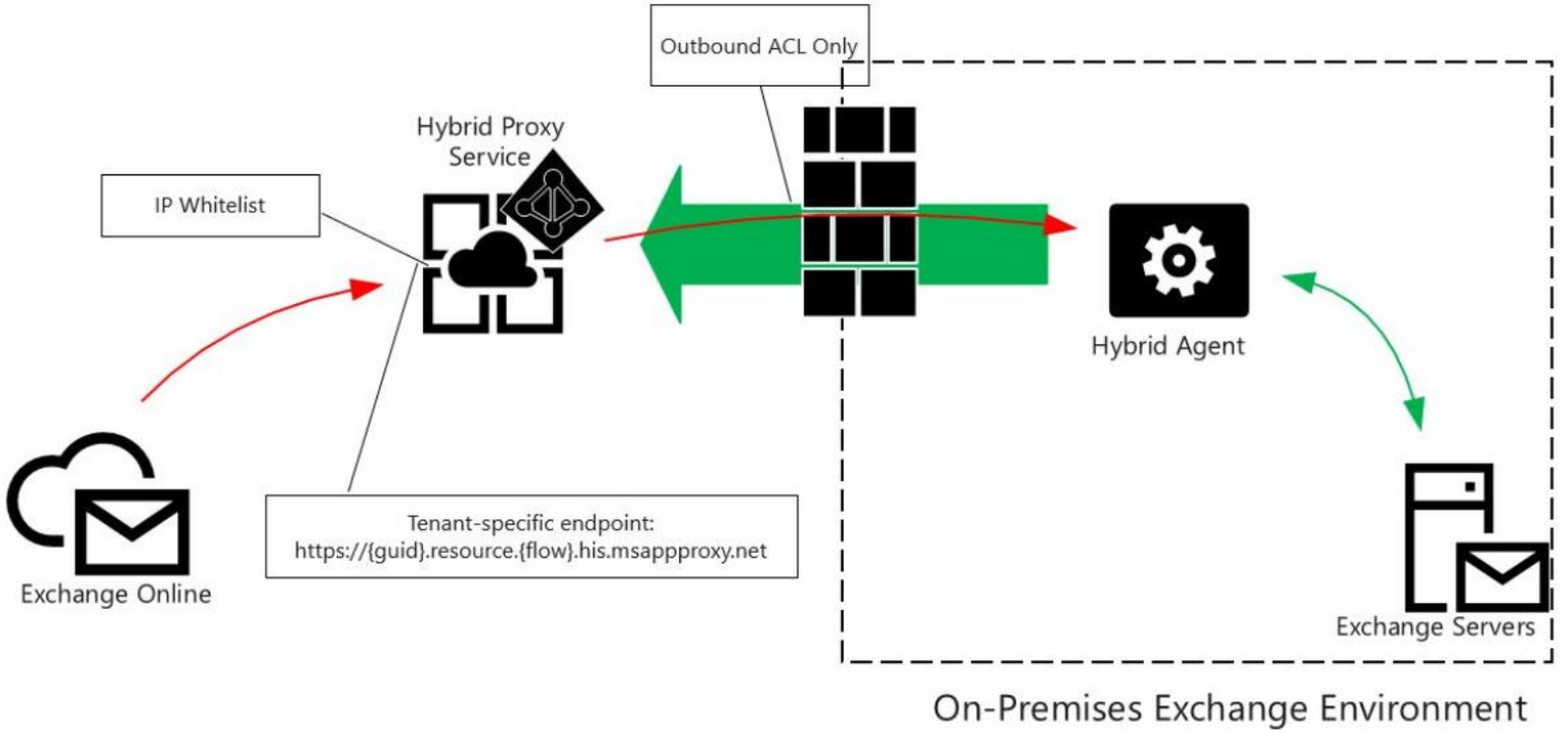
As the hybrid agent takes over the communication between Exchange Online and your Exchange environment, it provides the following advantages:

No custom DNS changes

No certificate changes

No firewall/network changes

Protect your Exchange servers



Review mail flow options for a hybrid deployment



Inbound email flow from the Internet

Inbound to on-premises Exchange Server

Inbound to Exchange Online

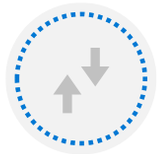


Outbound email flow to the Internet

Deliver Internet-bound messages directly

Route all Internet-bound messages through your on-premises Exchange servers

Route all Internet-bound messages through Exchange Online



Email flow between Exchange Online and the Exchange Server on-premises organization

Prepare to run the Hybrid Configuration Wizard



Before running the HCW, ensure the prerequisites are satisfied:

Deploy Exchange Server 2010 or later
Deploy Microsoft 365 for Enterprise or Business
Register custom domains
Deploy Azure AD Connect for AD synchronization
Configure AutoDiscover DNS records in non-Modern Hybrid deployments
Install Exchange services and assign them a valid digital certificate
Ensure mail flow is configured correctly to support hybrid mail flow
Configure Edge Synchronization (for Edge Transport) when applicable
Configure Unified Messaging-enabled (UM) mailboxes when applicable



Complete the following high-level steps to configure a hybrid deployment:

1. Sign up for Microsoft 365
2. Register your custom domains with Microsoft 365
3. Remediate identity within Active Directory
4. Deploy Azure AD Connect
5. Ensure Exchange Server is properly deployed

Examine hybrid deployment best practices



Use the Hybrid Configuration Wizard (HCW) app



Understand the requirements of Minimal Hybrid, Exchange Classic, and Exchange Modern Hybrid



Understand why your organization wants to implement a hybrid deployment



Start slowly, and then speed up when everything works



Do not change the MX resource record at first



Use multiple MRS Proxy endpoints where possible

Knowledge Check

Test your knowledge by answering the Knowledge Check questions at the end of this Learn module



Summary

This Module examined the following items related to planning and configuring a hybrid deployment using the Hybrid Configuration Wizard:



The Hybrid Configuration Wizard (HCW) is the tool used for setting up a hybrid configuration between an on-premises Exchange organization and Microsoft 365



The HCW includes a feature called Organization Configuration Transfer (OCT), which enables an organization to complete a one-time transfer of its organizational settings from its Exchange Server on-premises environment to Exchange Online



The Hybrid Agent connects to your Exchange Servers and the Hybrid Proxy Service located in your Microsoft 365 tenant
Any future requests from Exchange Online that require a connection to your Exchange Server will then be communicated using the Hybrid Agent



Complete the following high-level steps to configure a hybrid deployment:

1. Sign up for Microsoft 365
2. Register your custom domains with Microsoft 365
3. Remediate identity within Active Directory
4. Deploy Azure AD Connect
5. Ensure Exchange Server is properly deployed

Lab 7-Exercise 1 – Prepare Azure AD for a Hybrid Deployment

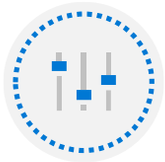


Lab exercises

Exercise 1: Prepare Azure AD for a Hybrid Deployment

Task 1

Configure your Tenant to Support Local Mail Transport



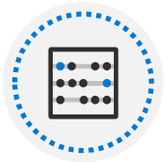
Task 2

Create a Custom Domain in Microsoft 365



Task 3

Configure the UPN Name for the Custom Domain



Task 4

Enable Exchange for the Custom Domain

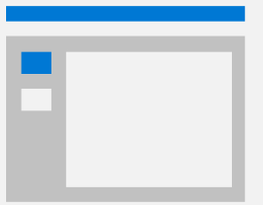


Task 5

Migrate On-premises user accounts to the Custom Domain



Module 3: Plan Mailbox Migrations



Introduction

This Module examines the importance of deciding the best migration option for your environment

After this Module, you'll be able to:

- Describe the migration and coexistence strategies with Exchange Online
- Identify the factors that help determine which migration option to use
- Identify when to change the DNS MS record to point to Exchange Online during a migration



Explore migration options to Exchange Online

Migration Approaches



Cutover Migrations



Staged Migration



Minimal Hybrid Configuration/
Express Migration



Full Hybrid Configuration



IMAP Migration



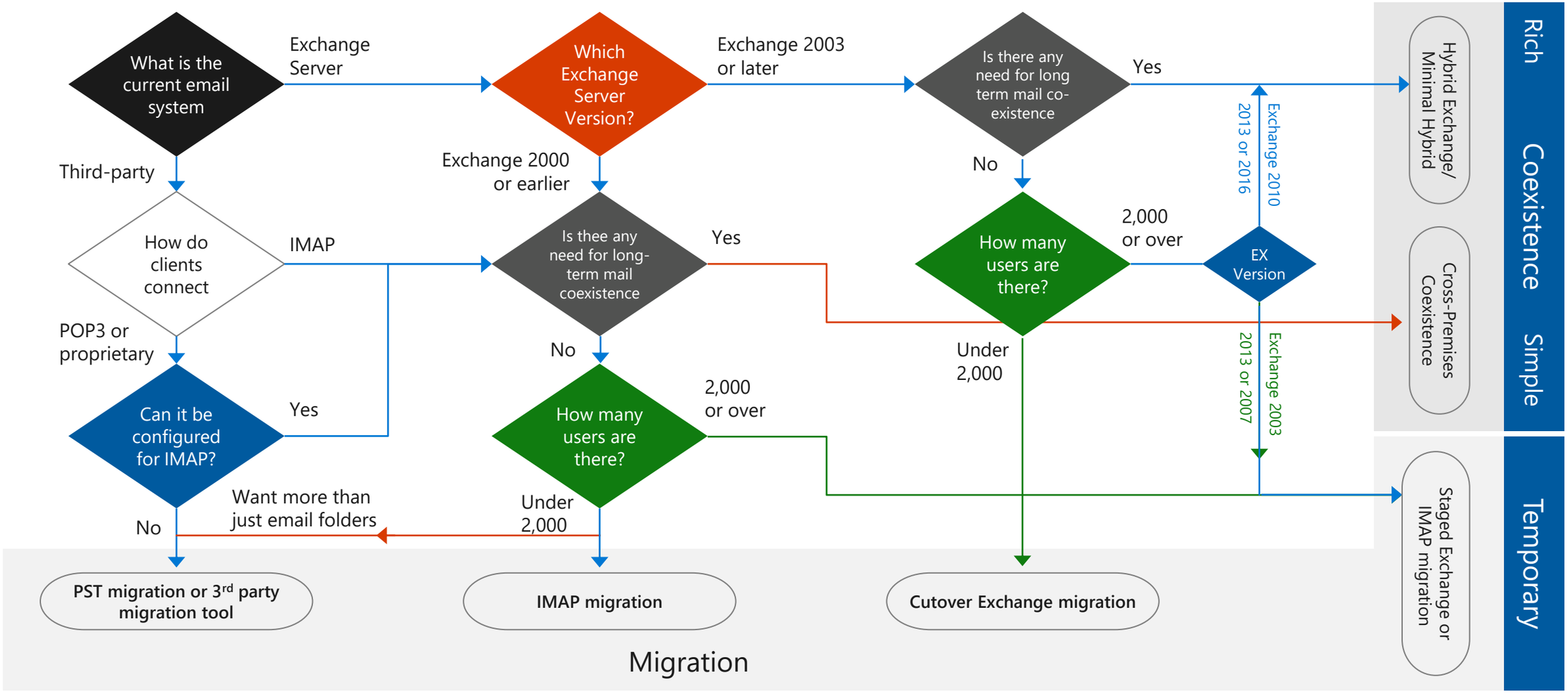
PST Migrations



Third-Party Migration

Determine which migration option to use

Exchange Online Migration Decision Matrix



Update DNS during a Microsoft 365 migration

The DNS MX record for a domain provides the main Internet SMTP message routing information and points to the target server for message delivery

You should consider multiple factors when deciding when to change your domain's DNS MX record to point to Exchange Online in a Microsoft 365 migration

These factors are dependent on each type of migration

Cutover Migration

Staged Migration

Full or Minimal Hybrid Configuration

IMAP Migration

Knowledge Check

Test your knowledge by answering the Knowledge Check questions at the end of this Learn module



Summary

This Module examined the following items related to planning mailbox migrations:



It's important to understand the difference between migration and coexistence

Migration - the process of moving user data from a source messaging system to Exchange Online

Coexistence - the process of maintaining mailboxes simultaneously in both your existing on-premises Exchange Server environment and Exchange Online; coexistence is also known as a hybrid configuration



The DNS MX record for a domain provides the main Internet SMTP message routing information and points to the target server for message delivery

You should consider multiple factors when deciding when to change your domain's DNS MX record to point to Exchange Online in a Microsoft 365 migration

Module 4: Run Mailbox Migrations



Module introduction

This Module examines how to plan for and run an IMAP migration, how to plan for a cross-tenant migration, and how to prepare source user objects for migration.

After this Module, you'll be able to:

- Describe the requirements for an IMAP migration and how it's carried out
- Move mailbox data using an IMAP migration
- Plan for a cross-tenant mailbox migration
- Prepare target user objects for a cross-tenant migration



Plan an IMAP migration



You need two things to migrate the contents of users' mailboxes from an IMAP messaging system to their Exchange Online mailboxes:

A CSV file that lists all the mailboxes to be migrated

The IMAP Migration wizard that’s located in the Exchange Admin Center



Limitations of the IMAP migration process:

You can only migrate message items in a user's mailbox; you can' migrate contacts, calendar items, or tasks

You can migrate a maximum of 500,000 mail items from a user’s mailbox

The maximum size of a single message item that can be migrated is 35 MB

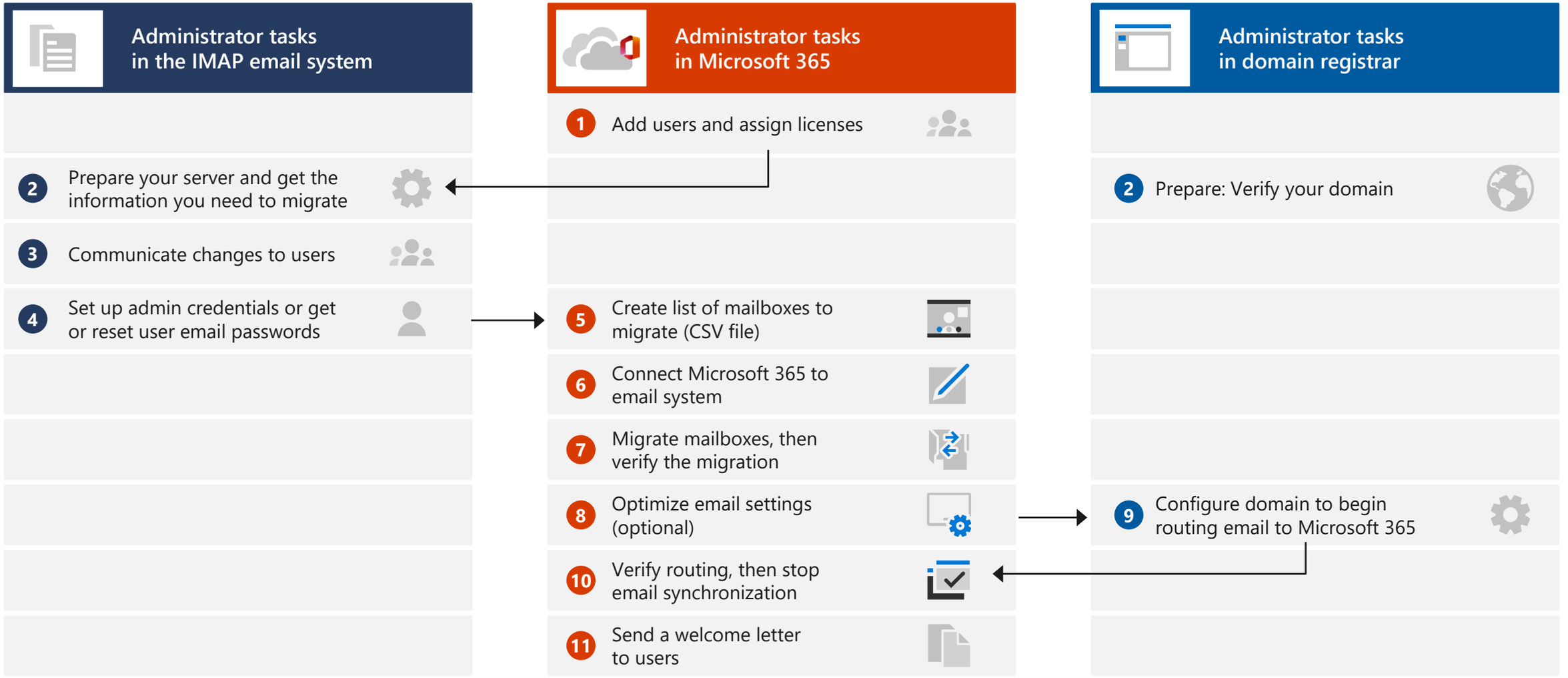


The effect of an IMAP migration on users:

If you can’t access user mailboxes, you’ll have to reset your users’ passwords

Since your users won’t know the new password, they'll be unable to access their old mailboxes during the email migration

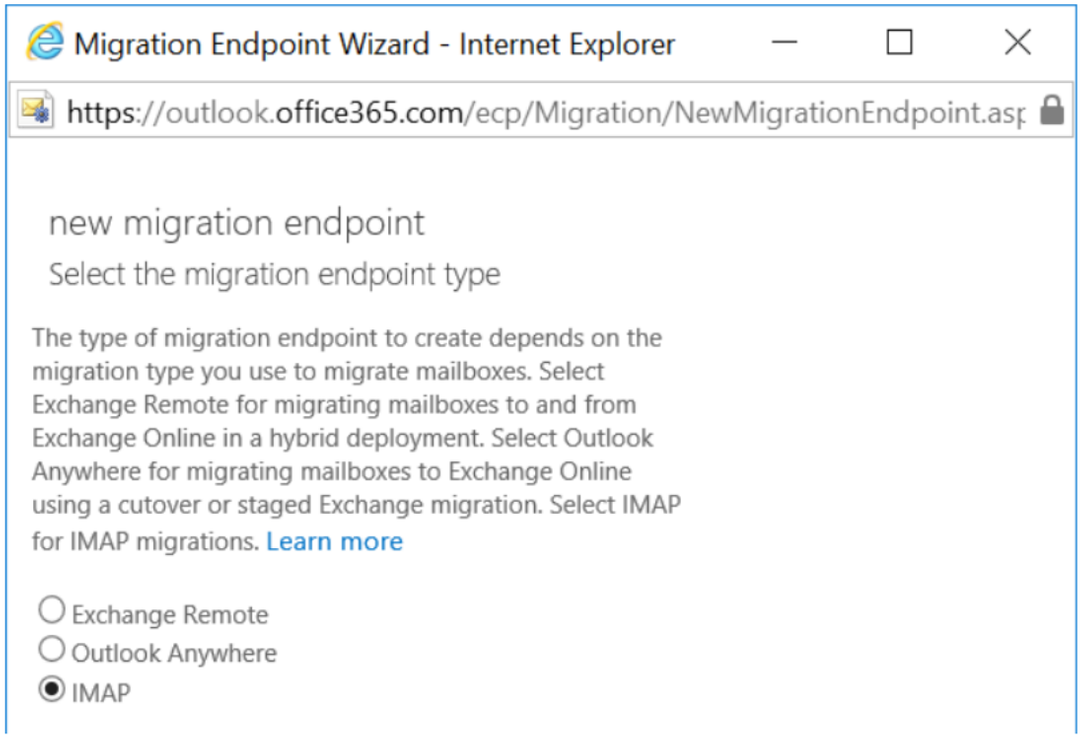
Run an IMAP migration



Move mailboxes using an IMAP migration

Steps for running your IMAP migration in the Exchange admin center:

1. Satisfy the migration prerequisites, such as:
   - Register any SMTP domains in Microsoft 365
   - Create all the target mailboxes in Exchange Online
   - Create a migration endpoint to connect Microsoft 365 to your email system
2. Create an IMAP migration batch and migrate your mailboxes
3. Stop the email synchronization and change the DNS MX record of all your domains to point to Microsoft 365
4. Allow either 72 hours for the final delta-synchronization to complete or manually start the migration batch again



Plan a cross-tenant migration



Cross-tenant migrations occur during mergers or divestitures when an organization must move its users' Exchange Online mailboxes into a new tenant

Cross-tenant Exchange mailbox migrations are done using Windows PowerShell

Cross-tenant migrations are started from the target tenant as migrations



Cross-tenant mailbox migration requirements:

Migrated users must be present in the target tenant's Exchange Online system as MailUsers, marked with specific attributes to enable the cross-tenant moves

You must have permissions to configure the Move Mailbox application in Azure, EXO Migration Endpoint, and the EXO Organization Relationship

At least one mail-enabled security group in the source tenant is required

Prepare the target (destination) tenant by creating the migration application and secret

Prepare the target tenant by creating the Exchange Online migration endpoint and organization relationship

Prepare the source tenant by accepting the migration application and configuring the organization relationship

Prepare source objects for a cross-tenant migration



Migrated users must be present in the target tenant's Exchange Online system as MailUsers, marked with specific attributes to enable the cross-tenant moves

For any mailbox moving from a source organization, you must configure a mail user object in the target organization

The system will fail moves for users that aren't properly configured in the target tenant



Users in the target organization must be licensed with appropriate Exchange Online subscriptions applicable for the organization

You may apply a license in advance of a mailbox move, but ONLY once the target MailUser is properly set up with an ExchangeGUID and proxy addresses

Applying a license before the ExchangeGUID is applied will result in a new mailbox provisioned in the target organization



You must ensure the target MailUser has no previous ExchangeGUID that doesn't match the Source ExchangeGUID

This situation can occur if the target mail end user was previously licensed for Exchange Online and provisioned a mailbox

If this situation occurs, you must clean up the cloud mail user using PowerShell to permanently clear the previous mailbox information

Knowledge Check

Test your knowledge by answering the Knowledge Check questions at the end of this Learn module



Summary

This Module examined the following items related to running mailbox migrations:



You need two things to migrate the contents of users' mailboxes from an IMAP messaging system to their Exchange Online mailboxes:

A CSV file that lists all the mailboxes to be migrated

The IMAP Migration wizard that’s located in the Exchange Admin Center



Completing an IMAP migration involves:

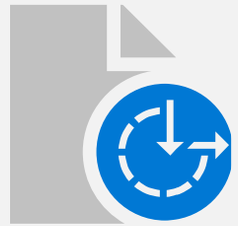
1. Satisfying the migration prerequisites
2. Creating an IMAP migration batch
3. Running the IMAP migration wizard



Cross-tenant migrations occur during mergers or divestitures when an organization must move its users' Exchange Online mailboxes into a new tenant

Cross-tenant Exchange mailbox migrations are done using Windows PowerShell

Lab 7-Exercise 2 – Configure your Hybrid Deployment



Lab exercises

Exercise 2: Configure your Hybrid Deployment

Task 1

Create Adatum's hybrid Exchange deployment



Task 2

Configure Mail Flow settings



Task 3

Create on-premises user mailboxes for testing



Task 4

Create a new Outbound Connector



Module 5: Troubleshoot Hybrid Deployments



Module overview

This Module introduces troubleshooting techniques for hybrid deployments

After this Module, you'll be able to troubleshoot:

- Directory synchronization
- Pass-Through Authentication and Single Sign-On
- Transport with Exchange Online
- Client Access in Coexistence
- Mailbox Replication Service



Troubleshoot Directory Synchronization

Once an organization establishes directory synchronization, it may occasionally experience issues with the synchronization or with individual objects syncing, such as:



Sync failed completely, which can be caused by:

The synchronization service stopped

Connector credentials are invalid

Configuration is set incorrectly



Single user object didn’t sync, which can be caused by:

Duplicate objects in Azure Active Directory

The object is filtered out

Invalid characters are present on the object

Troubleshoot Pass-Through Authentication and Single Sign-On

Users are quick to report when Azure AD Connect pass-through authentication (PTA) or single sign-on (SSO) aren't working or are in a degraded state



When pass-through authentication (PTA) isn't working, users are still prompted to input their credentials, but validation will fail. Troubleshooting includes:

If no users can sign in:

Verify the Microsoft Azure Active Directory Application Proxy Connector (WAPCSvc) is running
Verify Port 443 is enabled for outbound from the PTA server to Microsoft Azure

If one user is unable to sign in:

Review the sync logs to verify whether the user account synced to Azure AD
Verify the user account is enabled and not expired and the password has also not expired



When SSO isn't working, users must manually specify their password. Troubleshooting includes:

Verify the service URL (https://autologon.microsoftazuread-sso.com/) is defined in the Intranet zone
Verify client devices are joined and connected to the on-premises domain
Verify the SSO prerequisites are met
Verify that SSO is enabled

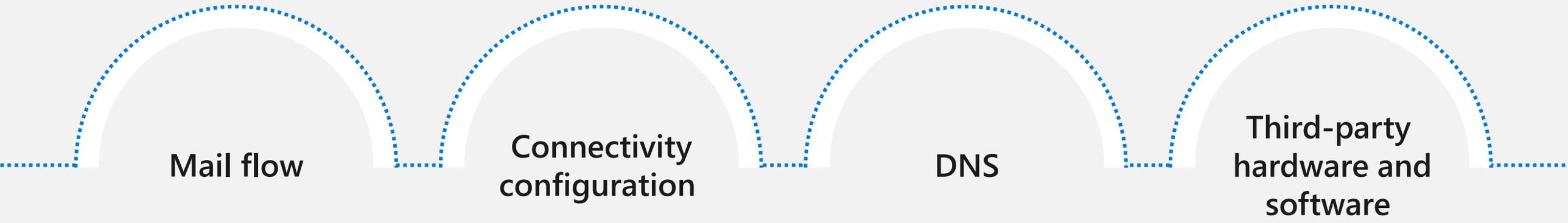


The Remote Connectivity Analyzer's Microsoft 365 Single Sign-On Test tool can help organizations troubleshoot SSO

Troubleshoot transport with Exchange Online

Having a thorough understanding of how mail transport works in a hybrid environment is helpful for troubleshooting

The key areas that affect transport in a hybrid environment include:



Troubleshoot client access in coexistence



When an organization's on-premises environment runs Hybrid coexistence, troubleshooting client connectivity becomes more complex

Not only must the organization manage its existing Exchange on-premises servers, but it must manage Exchange Online mailboxes



Verify the following items to prepare for client access in coexistence:

Firewalls – If an organization uses **Modern Hybrid**, firewall rules to support inbound Exchange Online connectivity aren't necessary; however, in all scenarios, outbound HTTPS connectivity from Exchange Servers is required

Certificates and DNS - If an organization uses **Classic Hybrid**, then it must use trusted, third-party Digital Certificates for HTTPS services, and the Digital Certificates must have Subject Alternative Names that match the configured Internal and External URLs and DNS names



Troubleshooting steps when client access issues occur in coexistence scenarios:

Run the Outlook Test Email AutoConfiguration tool

Test a new profile setup in Outlook

Run the Remote Connectivity Analyzer tool

Troubleshoot the Mailbox Replication Service



When you move mailboxes, public folders, and .PST files, the backend process that handles the data move is the Mailbox Replication Service (MRS)

Organizations typically troubleshoot MRS when they have moving issues, or when move performance is poor



Some indirect troubleshooting of MRS is often beneficial as well, such as collecting statistics to find information about move requests



Organizations can customize message throttling by adjusting settings in the MRS config file

Exchange Server uses dynamic message throttling, which can be disabled and replaced with customized throttling

The settings in the MRS config file are only used if dynamic throttling is disabled

If you change the settings in the MRS config file, you must restart the Microsoft Exchange Mailbox Replication service

Knowledge Check

Test your knowledge by answering the Knowledge Check questions at the end of this Learn module



Summary

This Module examined the following items related to troubleshooting hybrid deployments:



Messaging administrators must know how to troubleshoot and resolve common sync issues to maintain a healthy sync environment. Common issues include:

Sync fails completely - often the result of a backend configuration or communication problem, and less likely a problem with user or group objects

Single user object doesn't sync - When you have a functional sync, but you have a single object that doesn't sync, it typically means there's a problem with that object



When pass-through authentication isn't working, users are still prompted to input their credentials. However, there will often be a long pause during validation, and it will fail with an error message



When troubleshooting mail flow, you can start your troubleshooting at the edge (often Edge Transport servers or similar)

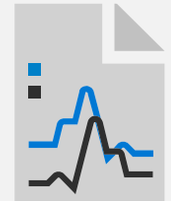


When an organization troubleshoots client access in coexistence, the problem is often limited to a single protocol



Organizations typically troubleshoot the Mailbox Replication Service (MRS) when they have moving issues, or when move performance is poor

Lab 7 – Exercises 3 – Test your Hybrid Deployment



Lab exercises (continued)

Exercise 3: Test your Hybrid Deployment

Task 1

Test the Hybrid Topology



Task 2

Migrate an On-Premises Mailbox to test your connectors

STOP! See note below



Task 3

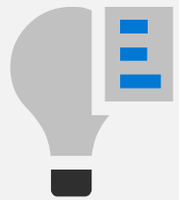
Test the newly migrated mailbox



IMPORTANT: Migrating an on-premises mailbox to Microsoft 365 in Exercise 2, Task 2 can take up to an hour to complete. You cannot test the newly migrated mailbox in Task 3 until the migration is complete.

To avoid sitting around for an hour, once you finish Task 2 perform the Paper-based lab exercise outlined in the next slide while the migration runs. Once the class has finished the paper-based lab, return to Exercise 2, Task 3 to test the newly migrated mailbox.

Exercise – Planning a Mail Migration to Microsoft 365



Paper-based lab exercise

Planning a Mail Migration to Microsoft 365

Fabrikam, Inc. is moving to Microsoft 365 as its cloud solution. The Enterprise Administrator has requested that you review the company's footprint and then develop a plan for migrating a group of on-premises mailboxes to Microsoft 365 as part of its mail migration pilot project. As part of that analysis, he has asked that you address some specific issues that concern him.

Download the planning document from GitHub – Located here

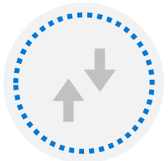
Step 1

Each student should review the information provided and answer the questions in the document



Step 2

After completing the assessment, exchange it with one of your fellow students and review each other's assessment. Identify the key differences between your two proposals



Step 3

Review your assessments with the class and come to a final solution for Fabrikam



Learning Path Review





Discussion – Learning Path review

What are your key takeaways from this Learning Path, and why?

What are the key features discussed in this Learning Path that you foresee implementing at your organization?

Microsoft 365

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